

LabelCheck

Automated Legal Metrology (Packaged Commodities) Rules, 2011 Compliance Screening & Multi-Angle Inspection Platform

Production URL:

<https://labelcheck-rho.vercel.app/>

AI Vision Engine:

Gemini 2.5 Flash (Multimodal)

Framework & UI:

Next.js 16 • React 19 • Tailwind

Cloud Persistence:

Supabase PostgreSQL + RLS

1 Executive Summary & Statutory Problem Statement

In India, pre-packaged retail commodities are governed by the **Legal Metrology (Packaged Commodities) Rules, 2011** enacted under the Legal Metrology Act, 2009. Manufacturers and packers are legally mandated to declare specific consumer protection information clearly on product packaging. In practice, non-compliance occurs frequently through:

- Omission of Mandatory Declarations:** Complete absence of MRP, metric net quantity, batch numbers, or manufacturing dates.
- Rule 9 Evasive Deferrals:** Printing placeholders on the main label (e.g. "MRP: See on side", "Mfg Date: See base") that redirect consumers away from the Principal Display Panel (PDP), violating Rule 9 which requires all declarations to appear on the principal panel.
- Pricing & Unit Sale Violations:** Non-declaration of Unit Sale Price (USP in ₹/g or ₹/ml) or non-inclusive MRP declarations.

LabelCheck solves this by providing automated, sub-2-second AI computer vision screening across single-photo and multi-angle packaging, detecting non-compliance, verifying mandatory citations, and cataloging deduplicated products into a centralized database.

2 Complete Technology Stack

⚡ Next.js 16 (App Router)

Turbopack compiler, React 19 Server/Client Components, dynamic API routes, and native App Router URL history for Android hardware back-button navigation.

🗿 Google Gemini 2.5 Flash

Interactions API with JSON Structured Output Schema and `thinkingBudget: 0` for sub-2s inference latency and precise field extraction.

🗄️ Supabase (PostgreSQL)

Cloud PostgreSQL database, strict Row Level Security (RLS) policies, Google OAuth 2.0 authentication, and private encrypted object storage.

🎨 Tailwind CSS & Framer Motion

Mobile-first dark-mode interface with fluid micro-interactions, responsive touch targets (≥48px), and full-width layout protection.

📷 Web MediaDevices API

In-page live camera viewfinder (`getUserMedia`) with camera flip, high-res canvas capture, and client-side Web Worker image compression.

🚀 Vercel Cloud Platform

Serverless edge execution, global CDN routing, automatic Git deployment, and dynamic proxy origin header resolution.

3 Dual-Scanning Modes

⚡ Mode 1: Quick Scan (Sub-2s Priority)

Single-photo upload/capture targeting the **3 highest-priority legal metrology declarations**:

- **MRP:** Maximum Retail Price in ₹ (inclusive of all taxes).
- **Net Quantity:** Standard metric units (g, kg, ml, l, count).
- **Month & Year:** Manufacturing/Packing/Import date.

🔍 Mode 2: Full Product Scan (10-Field Multi-Angle)

Multi-photo in-page camera session (up to 6 angles) capturing front, back, side panels, barcode, and batch stamps. Evaluates all **10 statutory Legal Metrology declarations** with parallel multi-angle reconciliation and product deduplication.

Core Design Philosophy: High speed for rapid field inspections, zero reliance on physical device caches, strict cloud data sync under authenticated Google credentials, and robust legal rule citations for every verdict.

4 End-to-End System Workings & Data Pipeline

The diagram below outlines the full lifecycle of a multi-angle inspection from mobile camera capture to AI analysis, conflict reconciliation, and cloud storage.

STEP 1

Multi-Angle In-Page Capture & Client-Side Compression: Inspector uses the live HTML5 camera (`MediaDevices.getUserMedia`) to capture 1 to 6 angles. Each photo is compressed client-side via Web Worker (1280px max dimension, JPEG quality 0.8) reducing payload size to <250KB per angle.

STEP 2

Parallel Background Inference: Captured photos are dispatched concurrently via `Promise.all` to `/api/analyze-full`. Gemini 2.5 Flash processes all angles in parallel with JSON schema enforcement, extracting text, confidence scores (0-100), and panel metadata.

STEP 3

Multi-Angle Reconciliation & Conflict Resolution: The `fullScanEngine.ts` consolidator evaluates all responses. For each of the 10 declarations, it reconciles detections across angles. If readings vary (e.g. Angle #1 reads ₹299, Angle #2 reads ₹299.00), it flags a soft conflict, adopts the highest confidence reading, and records the source photo index.

STEP 4

Rule 9 Deferral Detection: The AI inspects text for evasive phrases (e.g. "See on side", "Refer back panel"). Any deferred mandatory declaration is classified as **Not Detected** with a statutory citation explaining the Principal Display Panel violation.

STEP 5

Cloud Deduplication & Storage: The product is linked by Barcode or Brand+Commodity match. Photos are uploaded to Supabase Storage (`product-photos/{user_id}/{scan_id}/`) and records are inserted into `products` and `product_scans` tables under the user's Google User ID.

5 Legal Metrology Rule 9 Deferral Detection Logic

Under Rule 9 of the Legal Metrology (Packaged Commodities) Rules, 2011, all mandatory declarations must be printed directly on the principal display panel. Printing an evasive placeholder constitutes a statutory non-compliance infraction:

Observed Label Text	Assigned Status	Statutory Compliance Verdict & Legal Rule Reference
MRP: ₹ 499.00 (Incl. of all taxes)	Detected	Compliant under Rule 6(1)(e). Value clearly printed in INR with tax declaration.
MRP: See on Side Panel	Not Detected	Non-Compliant (Rule 9 Violation): Value deferred to another panel. Declarations must appear on the principal display panel.
Mfg. Date: Refer Base of Can	Not Detected	Non-Compliant (Rule 9 Violation): Manufacturing date deferred to base without direct declaration on label.
Country of Origin (Made in India)	Not Applicable	Domestic product. Mandatory country of origin declaration applies strictly to imported goods.
Blurry text: "Net Qty: 5..g"	Uncertain	Legibility warning. Confidence score downgraded (<60%) due to visual obstruction.

6 Product Deduplication & Hierarchy Engine

Deduplication Rule 1: Barcode Matching

When an EAN-13, UPC, or numeric barcode is OCR-detected, LabelCheck searches the user's catalog. If found, the new scan is automatically appended to the product's historical timeline.

Deduplication Rule 2: Brand & Commodity Matching

If no barcode is present, the engine searches existing products matching `Brand Name` and `Commodity Name`. If an ambiguous match is detected, the UI prompts the inspector to confirm or create a new variant.

7 The 10 Mandatory Legal Metrology Declarations Checklist

LabelCheck screens every retail product against the complete statutory checklist:

#	Field Identifier	Rule Citation	Plain-English Legal Requirement
1	mrp	Rule 6(1)(e)	Maximum Retail Price declared in Indian Rupees (₹), inclusive of all taxes.
2	netQuantity	Rule 6(1)(c)	Net weight, volume, or count in standard SI metric units (g, kg, ml, l).
3	mfgMonthYear	Rule 6(1)(d)	Month & Year of manufacture, packing, or import (e.g. "08/2026").
4	commodityName	Rule 6(1)(b)	Common or generic name of the commodity contained in the package.
5	manufacturerAddress	Rule 6(1)(a)	Complete name and physical street address of manufacturer, packer, or importer.
6	countryOfOrigin	Rule 6(1)(a)	Country of origin for imported goods (marked N/A for domestic products).
7	consumerCare	Rule 6(2)	Contact name, address, telephone number, and/or email for consumer complaints.
8	bestBeforeOrExpiry	Conditional	Best before, use by, or expiry date for food, cosmetics, and perishable items.
9	unitSalePrice	Conditional	Price per gram/ml/piece for packages containing quantity > 1 standard unit.
10	batchNumber	General	Batch, lot, or code number for supply chain traceability and recall enforcement.

8 Database Architecture & Security Schema

products (Catalog Table)

- id (uuid PK): Unique product identifier.
- user_id (uuid FK): Bound to auth.users.id.
- brand_name (text): Normalized brand name.
- commodity_name (text): Commodity category.
- barcode_number (text): EAN/UPC barcode.
- RLS: Users can only select/insert/modify their own products.

product_scans (Inspection Timeline)

- id (uuid PK): Unique inspection scan ID.
- product_id (uuid FK): References products(id).
- user_id (uuid FK): Bound to auth.users.id.
- checklist_results (jsonb): 10-field report & conflicts.
- photo_urls (text[]): Supabase storage photo URLs.
- RLS: Strict per-user ownership policy.

9 Mobile Navigation & Statutory Disclosure Architecture

4-Tab Fixed Bottom Bar (Mobile UX)

Fixed 4-tab bottom bar (**Scanner**, **History**, **Products**, **Profile**) with pb-36 viewport clearance ensuring action buttons ("Analyze All", "Sync Now", "Sign Out") are never obscured.

Android System Back Button Support

True Next.js App Router URLs (/scanner, /history, /products, /profile) creating native browser history entries without OAuth loopbacks.

Statutory Role Disclosure (Profile Tab): LabelCheck explicitly displays mandatory legal disclaimers stating that the application is an informational self-check screening tool and does not constitute certified government inspector credentials or official seizure memos.